Accessible Authentication

Inclusion Toolkit

What is it for?

When people can't use a login page, the harm can be serious. It can take away their right to privacy, security, and independence.

Logins should not rely on hard memory tests or puzzles, as these block users with cognitive needs. Forcing people to pass these checks often locks them out of vital services.

Logins designed with accessibility in mind are safer and easier to use for everyone.

How to support it

To make login pages accessible, developers should follow these steps:

- Don't rely solely on puzzles or memory tests. Always provide an alternative method
- Support password managers and allow users to paste text. This improves both accessibility and security
- Never disable autofill by setting autocomplete to 'off'. Forcing people to type complex details causes unnecessary difficulty
- Offer flexible alternatives like biometrics, one-time codes, and magic links

Useful Links

Understanding SC 3.3.8: Accessible Authentication (Minimum) (W3C)

Accessibility in Cybersecurity: A Practical Guide

The autocomplete attribute and web security (Mozilla Developer Network)

Accessible Links

Inclusion Toolkit

What is it for?

Links need descriptive text so everyone can understand them. While 'click here' links might work visually, people using assistive technology often scan links out of context and will just hear a repetitive list.

You should also avoid using raw uniform resource locators (URLs), as screen readers read out the whole address, including symbols, which is annoying and disorientating.

How to support it

Link text must clearly describe its destination without context. If it's inside a sentence, make sure it flows well while remaining informative.

You should always avoid vague phrases like 'here', 'click here', 'read more', and 'more information'.

Useful Links

Yale University - Links

WebAIM - Links and Hypertext

University of Reading: Accessible Links

Accessible PDFs

Inclusion Toolkit

What is it for?

PDF (Portable Document Format) is used to present all kinds of documents. Ideally, all the content in a PDF should be accessible to all. Here are some ways to make it accessible:

- Enable tagging, and set the title and language, so screen readers can navigate the content
- Design with high colour contrast, legible fonts, and a single-column layout including headers and footers
- Test the final document with a screen reader and a PDF Accessibility Checker

How to support it

Avoid PDFs whenever possible. Even ‘accessible’ ones aren't natively responsive for mobile and have major downsides. Instead, challenge the PDF requirement and choose an alternative:

- Semantic HTML web pages or web apps for the most accessible experience
- Word documents, Google Docs, or slide presentations, as they offer built-in accessibility checkers and excellent screen reader support
- ePub or Markdown formats to ensure reflowable text and clean semantic structures across all devices

Useful Links

Reading PDFs with reflow and accessibility features

Adobe: Creating accessible PDFs

Accessibility Debt

Inclusion Toolkit

What is it for?

Accessibility debt is like technical debt, but for accessibility. It refers to all the accessibility issues that build up over time when inclusive practices aren't followed from the start.

Just like technical debt, the longer it's left unmanaged, the harder (and more expensive) it becomes to fix.

How to support it

You don't choose to take on accessibility debt. It builds up when things like semantic HTML, alt text, proper labels, or keyboard navigation are skipped or broken. Here's how to manage it, or 'pay it off':

- Run regular accessibility audits (manual and automated)
- Add accessibility issues to your backlog and track them like bugs
- Prioritise fixes that block user journeys (for example, login, forms, navigation)
- Fix inaccessible components in your design system
- Include accessibility in your development and design workflow to prevent new debt

Useful Links

Understanding Accessibility Debt: What It Is and How to Manage It

Accessibility Debt Is Real — Here's How to Prevent It

Accessibility Debt: What Happens When You Do Accessibility Last

Accessibility Overlays

Inclusion Toolkit

What is it for?

Accessibility overlays are automated tools that promise a quick, effortless fix for web compliance.

While they detect basic errors like colour contrast, text size, and layout, they don't provide a complete solution. They can even introduce new barriers themselves, including inaccessible buttons and poor screen reader support.

To learn more, check out the Overlay Fact Sheet for a full explanation and feedback from real users.

How to support it

Don't use them. Make sure you're building accessibility into your designs and into your code, rather than relying on another company.

Useful Links

Siteimprove - Accessibility overlays

Overlay fact sheet

Accessibility Testing

Inclusion Toolkit

What is it for?

Accessibility Testing is the process of checking a digital product to ensure it can be used by people with disabilities. Its purpose is to find and fix barriers before your users do. Without testing, you're guessing if your product works for everyone. This can lead to significant problems, such as:

- People using screen readers or keyboards are blocked from completing tasks
- Users with disabilities face a broken or frustrating product
- Your product may fail to meet legal requirements based on Web Content Accessibility Guidelines (WCAG)
- Problems found after launch are much harder and more expensive to fix

How to support it

An effective strategy combines three testing methods:

- Automated software scans code for basic flaws, but it only finds about 30% of issues
- Manual testing relies on humans to check complex areas, including keyboard navigation, zoom levels, and screen reader support
- User testing gathers feedback from people with disabilities, which is the only way to prove a site is 'truly usable'

Useful Links

Best Practices for Accessibility Testing - A Comprehensive Guide

The Importance Of Manual Accessibility Testing

Accessibility Testing: How It Works and Why It's Important

Alt Text

Inclusion Toolkit

What is it for?

Some people cannot see the images we display. This includes people who are blind or have low vision. It also includes people with slow internet, little data, or low battery.

Using alternative text (alt text) ensures that everyone can understand the image content, even if they cannot see it. However, if an image is purely decorative and adds no extra information, give it an empty alt attribute (`alt=""`).

How to support it

- Write short, descriptive text that conveys the actual meaning and context of the image
- Avoid redundant phrases like ‘image of’, as screen readers already announce that
- Leave the description empty for purely decorative elements so assistive tools can skip them
- Plan for alt text during the design and content creation phases instead of at the end
- Use built-in accessibility checkers in software like Microsoft Office to catch missing descriptions

Useful Links

WebAIM Alternative Text

Write helpful Alt Text to describe images

What’s the alternative? How to write good alt text

ARIA

Inclusion Toolkit

What is it for?

Accessible Rich Internet Applications (ARIA) is a set of roles and attributes that make the web more accessible for users with disabilities. ARIA adds to HTML. It lets common web app interactions be passed to assistive technologies when there is no other way to do this.

Note: Not using ARIA is always better than using ARIA badly. Bad use of ARIA can make a website impossible to use for people who use assistive technology. ARIA attributes should only be used when semantic HTML can't do the job.

How to support it

- `aria-label` & `aria-labelledby`: Adds an accessible label to elements that do not have visible text. For example, a search button with only an icon should include an `aria-label` clarifying it. Use it on interactive elements or landmarks, not plain divs or text
- `aria-hidden`: Hides elements from screen readers without removing them visually. This is ideal for decorative or redundant elements
- ARIA roles add semantic meaning to content. These should be avoided when there is semantic HTML that can provide the same meaning (for example, don't use `role=listitem` and instead use an `<li>`)

Useful Links

- W3C - Rules of ARIA
- Mozilla: ARIA
- List of ARIA attributes
- Accessibility essentials every front-end developer should know
- What I Wish Someone Told Me When I Was Getting Into ARIA
- List of ARIA roles

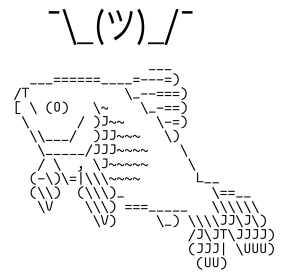
ASCII Art

Inclusion Toolkit

What is it for?

ASCII art is the practice of arranging punctuation and symbols to create a picture, like the ‘shrug’ or drawings of animals.

While sighted users see a picture, a screen reader reads the content in order, one character at a time. It cannot 'see' the image. Instead, the user hears a chaotic string of punctuation names: "Slash, backslash, underscore, vertical bar, colon..." It’s noisy and blocks them from getting to your actual content.



How to support it

You can still share these visuals, but you should change the format to be inclusive.

- Take a screenshot of the ASCII art you want to share
- Upload the screenshot rather than pasting the text directly
- Write a description of the image as alt text (for example, "A line drawing of a rabbit made of text characters")

Useful Links

How screen readers read special characters

RNIB - Making social media accessible

Assistive Technology (Beyond Screen Readers)

Inclusion Toolkit

What is it for?

Assistive technology includes tools that help people with disabilities use a computer.

Devices like magnifiers, switches, and voice control help people zoom in, or navigate without a mouse or keyboard.

Poor design, like tiny text or mouse-only buttons, can completely block these users.

How to support it

Support these tools by making website layouts compatible with them.

- Prioritise keyboard navigation: test with only ‘Tab’, ‘Shift + Tab’, ‘Enter’, ‘Space’, arrow keys and ‘Escape’
- Ensure layouts don't break when text is zoomed, and use clear, written labels instead of just icons
- Make sure any information shown on hover is also available by tabbing to it

Useful Links

WebAIM - Introduction to Assistive Technology

Basic Accessibility Testing

Inclusion Toolkit

What is it for?

Testing makes sure that what gets built works for everyone. The goal is to never release a beautiful page that's completely broken for someone using a screen reader or a keyboard.

While automated tools are useful, they can't find everything. They only catch about 30% of accessibility errors. An automated tool can check if a picture has an image description (alt text), but it can't tell you if the description makes sense. You still need to use your own judgement to check the rest.

How to support it

You don't need to be an expert to test your site. Try these three quick checks right now:

- Use 'Tab' and 'Shift + Tab' to move, 'Enter' or 'Space' to select, arrow keys in menus, 'Escape' to close. Can you reach every button?
- Zoom your browser to 200% using the Control (Ctrl) or Command (Cmd) key and '+'. Ensure text doesn't overlap or vanish off the screen
- Run a free scanner like Lighthouse or Axe DevTools in Chrome. It'll quickly point out basic bugs, like poor colour contrast

Useful Links

W3C - Easy Checks (A first review of Web Accessibility)

WebAIM - Testing with a Keyboard

Breadcrumbs & Wayfinding

Inclusion Toolkit

What is it for?

Breadcrumbs show a user's current location within a website's hierarchy.

They help everyone understand where they are, where the page sits in the site, and how to navigate back. This context is essential for screen reader users and people with cognitive impairments.

Bad wayfinding leaves users confused, disoriented, and forced to start their journey over.

How to support it

To make breadcrumbs accessible, you should build them using structured code.

- Wrap an ordered list in a navigation element with a 'breadcrumb' label
- Don't make the final item a link and ensure you mark it as the current page
- Hide visual elements like arrows or slashes from screen readers or use a pre-built accessible component

Useful Links

Breadcrumbs for Accessibility: Examples & Best Practices

GOV.UK Design System: Breadcrumbs

The Admin Bar: Accessible Breadcrumbs

Buttons

Inclusion Toolkit

What is it for?

Buttons let people take an action, such as submitting a form, saving information, or opening a menu. If a button isn't accessible, people can be completely blocked from finishing a task.

Accessible buttons help people who use a keyboard, a screen reader, voice control, or a switch device, as well as people with low vision or motor impairments.

How to support it

- Use a button for actions, and a link to go to a new page
- Use the native `<button>` element, so it works with ‘Tab’, ‘Enter’ and ‘Space’ keys
- Give every button a clear label that describes the action, such as 'Save and continue'
- Give icon-only buttons an accessible name, such as an `aria-label`
- Make sure buttons have a visible focus state, good colour contrast and a large enough tap target
- Avoid disabled buttons, and don't rely on colour alone to show a warning

Useful Links

GOV.UK Design System: Button

NZ Government Web Accessibility Guide: Buttons for keyboard, mouse and touch

W3C WAI: Button examples (ARIA Authoring Practices Guide)

Captions & Transcripts

Inclusion Toolkit

What is it for?

Captions and transcripts textually represent audio content to make media accessible.

Captions are synced with video, making them vital for users who are deaf, hard of hearing, or have cognitive difficulties.

Transcripts provide an untimed text alternative, which helps people who learn best by reading.

How to support it

Many standard platforms make it easy to generate and display these text alternatives.

YouTube and video call tools like Zoom, Microsoft Teams, and Google Meet have built-in captions. Many also give you a transcript of a recording.

Tools like YouTube, Microsoft Teams or Slack can auto-generate a transcript when you upload a video. Simply copy the text, fix any odd punctuation, and share the final document.

Useful Links

W3C WAI: Transcripts

What are Captions?

YouTube: Add subtitles and captions

How to Create a Transcript: A Step-by-Step Guide (2025)

Carousels & Sliders

Inclusion Toolkit

What is it for?

A carousel is a sliding banner that automatically cycles through images or messages on a webpage, like a digital billboard.

When content moves on its own, people who read slowly or have difficulty focusing can miss it entirely. People who use special software to read screens aloud often cannot follow the content in the right order. And people who cannot use a mouse frequently have no way to control the carousel at all.

Research shows that most people ignore carousels and never click past the first slide.

How to support it

The best solution is often to not use a carousel. Showing content in a simple list or grid means everyone can see it, without any of the accessibility problems.

When a carousel is truly needed, it should never move on its own without the user choosing to move it on. The controls need to be big, clear, and work with a keyboard as well as a mouse. And nothing important should only appear inside a carousel.

Useful Links

Should I Use a Carousel? (shouldiuseacarousel.com)

W3C Carousel Accessibility Tutorial

SC 2.2.2: Pause, Stop, Hide (Level A)

Colour Blindness

Inclusion Toolkit

What is it for?

Colour blindness affects many people, altering how they perceive the world around them.

It's a huge issue if colour is the only way information is conveyed.

Common indicators like 'red for error' and 'green for success' can look the same to some users.

How to support it

Design content to accommodate the various ways that people see.

Different conditions mean people can't always distinguish between reds, greens, blues, or yellows.

Don't rely on colour alone, and always include a text label, an icon, or a pattern.

Useful Links

Coblis - Color Blindness Simulator

We are Colorblind

W3C - Use of Color

Colour-blind Series: Protanopia

Inclusion Toolkit

What is it for?

Protanopia is a form of colour blindness that affects the ability to see red. People with protanopia cannot distinguish between red and green, and red tones can appear very dark or almost black. It's one of two types of colour blindness that belong to the 'red-green colour blind' group.

This matters in digital products, because colour is often used to share important information. A form that highlights errors in red, or a chart using red and green bars side by side, becomes very difficult to interpret.

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)

Colour-blind Series: Deuteranopia

Inclusion Toolkit

What is it for?

Deuteranopia is a form of colour blindness that affects the ability to see green. Like Protanopia, it's a type of 'red-green colour blindness'. Protanopia affects the cells that sense red. Deuteranopia affects the cells that sense green. It's the more common of the two, affecting about 1% of men. It's very rare in women.

In practice, the experience is similar. Red and green become very difficult to tell apart, making colour-coded information unreliable. A traffic light style status indicator, a heatmap, or a form validation message relying on red and green will all present the same challenge.

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)

Colour-blind Series: Tritanopia

Inclusion Toolkit

What is it for?

Tritanopia is a form of colour blindness that affects the ability to see blue. Protanopia and deuteranopia are both types of "red-green colour blindness". Tritanopia is different: it affects blue and yellow. People with tritanopia struggle to distinguish blue from green, and yellow from pink or red. It is significantly rarer than the red-green types, affecting less than 0.01% of the population.

In practice, this means colour choices that feel safe for red-green colour blindness can still cause problems. A design that uses blue and green to tell groups apart can be very hard to understand. So can yellow text on white, or white text on yellow.

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)

Colour-blind Series: Anomalous Trichromacy

Inclusion Toolkit

What is it for?

All three types of colour sensing cells are present, but one of them doesn't function as strongly as it should. The result is weakened rather than absent colour vision.

- **Protanomaly** is a reduced sensitivity to red, making it harder to distinguish red from green or brown
- **Deuteranomaly** is a reduced sensitivity to green. It is the most common form of colour vision deficiency overall
- **Tritanomaly** is a reduced sensitivity to blue, making blue and green, and yellow and red, harder to tell apart

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)

Colour-blind Series: The Rarer Types

Inclusion Toolkit

What is it for?

Achromatopsia is complete colour blindness, meaning a person sees the world entirely in greyscale. It's very rare. People with it are often sensitive to light and cannot see fine detail clearly.

Achromatopsia is one type of monochromacy. Another is blue cone monochromacy, where only one type of colour sensing cell works at all. This leaves very little colour vision.

How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use a colour blind simulation tool, such as Web Disability Simulator for Chrome, to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (for example, by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)

Colour Contrast

Inclusion Toolkit

What is it for?

Websites, social media images, posters, billboards, and other outdoor advertising should have a good colour contrast between text and background colour. That includes links, icons, buttons, and any other information.

It's useful for people with vision impairments, colour vision deficiency, cataracts, elderly people, and people in different lighting situations (for example, sun glare on a phone screen).

The general rule is that text and images of text must have a contrast ratio of at least 4.5:1. Large text, which is 18 point or bigger, or 14 point if bold, can have a contrast ratio of 3:1.

How to support it

Choose colours with sufficient contrast. There are lots of colour contrast checkers online, including Colour Contrast checker (colourcontrast.cc), WebAIM's contrast checker, Coolors contrast checker.

Enter your background colour and foreground colour. You want to make sure that it gives you a contrast ratio that is 4.5:1 or higher (so 6:1 is good, 2:1 is bad).

Useful Links

Colour contrast - Why does it matter?

Accessible use of colour

WCAG 2.1 - Contrast (minimum)

Consistency

Inclusion Toolkit

What is it for?

When navigation, buttons, and labels behave consistently across every page, users can build a mental map of the product and move around it confidently.

This matters most for people with cognitive or learning disabilities, memory difficulties, ADHD, anxiety or dyslexia.

How to support it

- Designers can ensure the same components are used for the same actions across every screen
- Content writers can use the same labels for the same things throughout a product. Calling something 'Sign in' on one page and 'Log in' on another creates unnecessary confusion
- Developers can build from a shared component library, so the same element always looks and behaves the same way

Small inconsistencies that feel trivial to one person can be a significant barrier to another.

Useful Links

Understanding SC 3.2.3: Consistent Navigation (Level AA)

Making Content Usable for People with Cognitive and Learning Disabilities

Dark Mode

Inclusion Toolkit

What is it for?

Dark mode swaps the typical view to display light text on a dark background, toning down screen brightness.

It can reduce eye strain, save battery on certain screens, and help users with light sensitivity.

However, it isn't a perfect fix, as it can cause text to blur for users with astigmatism, and it may confuse certain types of colour blindness.

How to support it

You can implement dark mode through coding preferences or manual user options. The best option is always to give your user the choice.

Use the `prefers-color-scheme` media query to match system settings, and add a toggle switch so users have full control.

Designing for email is much more complex, as different clients render dark mode differently, some inverting colours or applying their own styles.

Useful Links

- Dark mode accessibility myth
- Pros and cons of dark mode
- Designing accessible dark themes
- 5 dark mode myths and realities

Device Orientation

Inclusion Toolkit

What is it for?

Think about how you hold your phone. Sometimes you hold it tall (portrait) to read, and sometimes you turn it sideways (landscape) to watch a video.

For some people, turning their device is not an option. A person who uses a power wheelchair might have their tablet bolted to the armrest, so it won't fall. If a website forces the screen to be held portrait, but their tablet is locked to landscape, they can't use it. It's like bolting a TV in portrait orientation on your wall and being told to tilt your head to watch it.

How to support it

Websites and apps should work no matter how the screen is turned
Never use code that forces the screen to stay in just one direction
Build designs that stretch or stack neatly to fit the screen, whether it’s tall or wide

Useful Links

W3C - Understanding Orientation
A11y Tip: Device Orientation

Error Messages & Validation

Inclusion Toolkit

What is it for?

Form errors are a major barrier for all users, but they can completely block users with disabilities. Inaccessible errors include:

- Just highlighting a field in red is useless for screen reader users and people with colour blindness
- Vague messages like 'Invalid input' or 'Error' tell the user nothing about how to fix it
- When a form fails to submit, screen reader users are often left at the top of the page, not knowing what or where the error is

How to support it

- Be specific. Don't say 'Invalid', instead say something like 'Please enter a valid email address'
- Don't use colour alone. Always add a text message, an accessible icon, or both, next to the field
- Connect the error. Use `aria-describedby` to link the input field to its error message
- Manage focus. If a form fails to submit, move focus to the first field with an error
- Use `aria-invalid="true"` on the field so assistive technology knows it has an error

Useful Links

W3C WAI - Validating Input

Level Access - Designer Tips: Creating Accessible Forms

Flashing & Photosensitivity

Inclusion Toolkit

What is it for?

High-energy videos with strobing effects or rapid transitions can be physically dangerous. For people with epilepsy, if content flashes more than three times per second, it can trigger seizures. For people with vestibular disorders (inner-ear balance issues), intense motion like fast spinning or "shaky cam" can cause instant dizziness, sickness, and vertigo.

How to support it

Safety is about moderation.

- Ensure nothing flashes more than three times in one second
- Avoid rapid spinning or intense zooming
- If a video must contain flashing (for example, concert footage), add a static "Warning: Flashing images" cover slide so users can scroll past before it auto-plays
- Avoid autoplay on websites; let users choose to start the video manually

Useful Links

Epilepsy Society - Photosensitive Epilepsy

Designing Safer Web Animation For Motion Sensitivity

Focus States

Inclusion Toolkit

What is it for?

Focus states occur when we focus on an interactive element. Interactive elements include links, buttons, and input fields. You can focus on an element by clicking it (if the click doesn't take you somewhere else), or by pressing the ‘Tab’ key until the element is selected.

Focus states have their own Web Content Accessibility Guidelines (WCAG) rules. The focus indicator must be visible, have a contrast ratio of at least 3:1 against nearby colours, and not be completely hidden by sticky headers or other content.

How to support it

If you're building or designing a new component, make sure its focus state meets all three of these rules.

Useful Links

Understanding focus state styles
Wise - Focus states
Focusing on focus states

Focus Traps

Inclusion Toolkit

What is it for?

A focus trap keeps keyboard focus inside one part of the page while it's open, such as a modal. Pressing Tab only moves between the items inside it, so the user doesn't get lost on the page behind it. A good focus trap always lets the user leave, for example by pressing Escape or using a close button.

How to support it

Only use a focus trap where it's needed, such as in a modal. Unwanted traps can be added by accident. Always test your pages using only the keyboard before any changes go live.

If you genuinely want a focus trap, use a pre-built accessible component like a Modal, which handles focus trap behaviour automatically.

Useful Links

A Primer on Focus Trapping

How to Build Accessible Modals with Focus Traps

Fonts

Inclusion Toolkit

What is it for?

Some fonts are much easier to read than others. However, some fonts may appear to be easy to read to some, but very challenging for others.

Fonts matter to everyone, but especially to the following people:

- People with reading difficulties (like dyslexia)
- People with visual impairments

How to support it

Mirroring is when a reader mixes up two characters, because one looks like the other turned around (for example, `b` and `d`). Transposition is when letters seem to swap places (for example, “loin” read as “lion”).

To help, choose a font where pairs like `q p, d b, 0 O, n u, i l 1 I, 6 9, a 8, a 6, 6 g 9` and `rn m` look clearly different from each other.

Fonts such as Atkinson Hyperlegible Next have become popular with individuals who have visual or cognitive disabilities.

Useful Links

Introducing accessibility in typography

Siteimprove: Accessible fonts

What is an accessible font?

Forms

Inclusion Toolkit

What is it for?

An accessible form is an online form that everyone can fill in, including people with vision, movement, or cognitive disabilities. Inaccessible forms create significant barriers, including:

- Missing or unlinked labels
- Poor keyboard navigation
- Vague error messages
- Small tap targets
- Overly complex layouts or time limits

How to support it

- Always use `<label>` tags and associate them with their inputs
- Use a `<fieldset>` to group related items and a `<legend>` to describe the group
- Provide clear error validation that identifies the field, describes the problem, and suggests a correction
- Ensure every form field, link, and button can be reached using only the keyboard, in a logical sequence
- Use high-contrast colours, and ensure all clickable elements have a large enough target size

Useful Links

W3C WAI: Forms Tutorial

WebAIM: Creating Accessible Forms

MDN: Web Forms

Good Alt Text

Inclusion Toolkit

What is it for?

While basic alt text just labels an image, good alt text explains its purpose. It helps people understand why the image is there, not just what it shows.

- Delivers the exact message, data, or emotion the image is meant to convey
- Provides an equivalent user experience instead of just following a rule
- Prevents confusion caused by vague descriptions like ‘chart’ or ‘screenshot’
- Focuses only on the details that matter to the surrounding text

How to support it

Supporting good alt text means moving past basic labels and focusing on the specific message your image is meant to communicate.

- Think about what information a user misses if the image fails to load
- Tailor the text to match the specific context of the surrounding page
- Avoid using alt text like ‘image.png’ or ‘graph’, which give the user no useful context
- Use good alt text like ‘Line graph showing a 20 per cent increase in app sign-ups this quarter’ to provide immediate value

Useful Links

Write helpful Alt Text to describe images

Good Alt Text, Bad Alt Text — Making Your Content Perceivable

What’s the alternative? How to write good alt text

Head Pointer

Inclusion Toolkit

What is it for?

Head pointer allows users with limited hand function to control a computer cursor using head movements and, in some cases, facial expressions. It's an accessibility feature available on macOS and can be enabled through System Settings.

How to use it

As Head Pointer is so customisable, a full breakdown isn't possible here. However, the short version is that it can control your cursor by tracking your head movements, and it can use facial expressions to perform certain actions. For example, you can set it up to left-click when you stick your tongue out, or when you wink.

Look at the links in the Useful links section, get it set up to your preferences, and try working with it!

Useful Links

Move the pointer using Head Pointer on macOS

An introduction to macOS Head Pointer

Headings

Inclusion Toolkit

What is it for?

Headings are important for more than just making text stand out. Headings are especially important for the following:

- People who use screen readers
- People with cognitive difficulties
- Search engines

How to support it

Headings shouldn't be used for styling. They tell a screen reader that your text is a heading.

- There should only be one title (or `h1`) per document, slide, or web page
- Headings should follow in a sequential order. You can return to previous heading levels, but you should never skip a heading level on your way down (so `h4` to `h2` is perfectly fine, `h2` to `h4` is not)
- If you're using headings incorrectly, search engines will get the wrong information, and search engine optimisation (SEO) can suffer as a result

Useful Links

How to: Accessible heading structure

Arizona State University: Accessible headings

W3C WAI: Headings tutorial

Home Office Design System: Headings

Hover States

Inclusion Toolkit

What is it for?

Hover states are what happen when a mouse pointer moves over an interactive element. Interactive elements include things like links, buttons, and inputs.

Hover states must also follow all the usual accessibility guidelines. For instance, content that appears on hover or focus must stay visible when the pointer moves onto it, and Escape must close it.

How to support it

If you're building or designing a new component, make sure it has hover states that follow all accessibility guidelines (for example, colour contrast, target size, descriptive text, and so on)

Useful Links

A whole lot of bovver over hover: problems with content that shows on hover

Accessible button design

Hover Actions and Accessibility: Addressing a Common WCAG Violation

Inclusive Language

Inclusion Toolkit

What is it for?

Inclusive language means using words that respect everyone. It avoids words that leave out, sideline, or put down groups of people, for example, because of disability, race, gender, sexuality, age, or wealth.

- Ableist language (for example, 'lame', 'crazy', 'blind spot') uses disability as a negative metaphor
- Biased or gendered terms (for example, 'salesman', 'guys') can exclude people
- Overly complex jargon or idioms can be significant barriers
- Using labels (for example, 'the disabled') defines people by their condition

How to support it

- Use person-first language (for example, 'a person with vision loss') unless people prefer identity-first (for example, 'an autistic person'), as the Deaf community generally does
- Avoid ableist metaphors. Instead of 'fell on deaf ears', say 'the feedback was ignored'
- Use neutral terms like 'everyone', 'team', or 'you all' instead of 'guys'
- Be specific and clear. Avoid jargon and write in plain language

Useful Links

Home Office Design System: Inclusive language

Harder than you think: developing an inclusive language guide

The Quick Guide to Inclusive Language

Keyboard Only

Inclusion Toolkit

What is it for?

Keyboard only is a method for navigating the web using just a keyboard (not using the mouse at all). Many people with disabilities navigate using only the keyboard, but some users without disabilities also navigate using the keyboard for speed and efficiency.

It's especially important for people with motor skills disabilities, people who struggle with fine motor control, people with tremors, people with little to no use of their hands, and people with visual impairments.

How to support it

Make sure that all interactive elements (buttons, links, and so on) can be navigated to using the keyboard, and that all interactivity still functions as expected.

The focus should also be clear: if your focus is on a button, can you tell that's where your focus is just by looking?

To test, press `Tab` to move forward, `Shift` + `Tab` to go back, arrow keys in menus, `Enter` or `Space` to click.

Useful Links

- WebAIM: Keyboard accessibility
- MDN: Keyboard accessible
- How to Enhance Accessibility with Keyboard-only Navigation

Landmarks & Roles

Inclusion Toolkit

What is it for?

Imagine trying to navigate a city with no street signs, where every building looks the same. That is what a website feels like to a screen reader user without Landmarks and Roles.

- Landmarks: These are like zones on a map. They tell the user 'This is the Navigation' or 'This is the Main Content'
- Roles: These are like name tags. They tell the user exactly what an item is

How to support it

The best way is to use semantic HTML (standard HTML tags). When you use the right tag, the landmark or role is added for you automatically.

- Landmarks: Use `<nav>` for menus and `<main>` for your main content
- Roles: Use `<button>` for buttons and `<a>` for links

You can use code to force a plain box (`<div>`) to act like a `<button>`, but it's risky and takes much more work.

Useful Links

CSS Tricks - Why, How, and When to Use Semantic HTML and ARIA

HTML Semantics and Accessibility Cheat Sheet

How to Improve Web Accessibility with Landmarks

Link Text

Inclusion Toolkit

What is it for?

Many screen reader users navigate websites by opening a list of only the links on a page. If your links just say 'Click here' or 'Read more', their list will just read out, 'Click here, Click here, Read more'. That doesn't help them find what they are looking for. Good link text makes sense all by itself, even if you hide the rest of the page.

Good link text doesn't need any special code. It's all about writing clear, helpful words, and everyone benefits from that, not just screen reader users.

How to support it

Write links that tell the user exactly where they are going, what they're doing, or what they are opening

- Be descriptive. Use 'Read our 2024 report', not 'To see our new report, click here'
- Avoid raw uniform resource locators (URLs). Screen readers will read out the whole address, including slashes and symbols

Users jumping from link to link do not hear the surrounding text. They only hear the words that you made clickable

Useful Links

Writing Meaningful Link Text
WebAIM: Links and Hypertext
W3C: Link Purpose

Meetings

Inclusion Toolkit

What is it for?

Meetings are part of most workdays, and there are many ways to make them fully accessible and inclusive. This helps everyone, but it can especially help:

- Neurodivergent people
- People with anxiety
- People with learning disabilities

How to support it

- Use an inclusive meeting invite template that asks about accessibility requirements
- Give a short visual description of yourself
- Let everyone know your pronouns
- Shorten your meetings by 5-10 minutes
- Allow for people to move, doodle, fidget, or stand during the meeting
- Let your attendees know that they can turn on closed captions or the meeting transcript
- Include breaks for meetings longer than an hour

Useful Links

University of Reading: Accessible meetings

Top tips for hosting accessible and inclusive online events

Accessibility tips for inclusive Microsoft Teams meetings and live events

Modals

Inclusion Toolkit

What is it for?

Modals are manually triggered pop-ups that provide extra information. For example, a shop might use a modal to explain what each condition grade of a refurbished phone means.

Modals make it possible to include all the required information on a page, without presenting users with a giant page full of text.

Modals are very useful for (but not limited to) people with cognitive impairments, people with ADHD, people with autism and screen reader users.

How to support it

In general, modals can be tricky to build accessibly. Important considerations include the user's focus and keyboard navigation.

- Move focus into the modal when it opens, and keep it there while it's open
- Let users close it with the Escape key as well as a visible close button
- Return focus to the element that opened it when it closes

Useful Links

Accessibility Essentials - Modals

Neurodiversity & Inclusive Design

Inclusion Toolkit

What is it for?

Neurodiversity describes the natural variations in how human brains function, process information, learn, and interact. Bad design can create problems for thinking and processing, such as:

- Cognitive overload caused by busy layouts, auto-playing videos, and flashing animations
- Using jargon, complex words, or vague phrases
- Menus, buttons, or layouts that change from page to page
- Long blocks of text, and unnecessary urgency

How to support it

- Keep it simple and clear. Use plain English and short sentences
- Make it predictable. Use headings and lists to organise content
- Reduce distractions. Let the user be in control. Avoid auto-playing videos or flashing animations
- Be forgiving. Write error messages that clearly explain what is wrong and how to fix it
- Make content easy to read by breaking up long text with headings, lists, and empty space

Useful Links

Neurodiversity and designing for difference

Scott Logic: Designing for Neurodiversity

Personas

Inclusion Toolkit

What is it for?

Essentially, Personas are fake people. Their lives, issues, and accessibility requirements allow you to role-play as that person while you test for accessibility. This brings a human side to accessibility testing. They aren't real, but real people live with the same conditions and needs.

How to support it

Start by going to the HMRC Personas website, and run through each of their Personas, attempting the tasks that are set. Once you've done that, take what you've learnt and the methods that you've used, and use them to test your own work for good accessibility practices.

Useful Links

HMRC Personas

Prefers-reduced-motion

Inclusion Toolkit

What is it for?

Some people struggle with motion on websites. This can be due to several reasons, including ADHD, motion sickness and vestibular disorders. Less motion also saves battery and data.

Use the `prefers-reduced-motion` media query to stop motion for people who choose not to see it.

Mac: `System Settings`, `Accessibility`, `Display`, `Reduce motion`

Windows: `Settings`, `Accessibility`, `Visual effects`, `Animation effects`

How to support it

It's like any other media query. The two options available are `reduce` or `no-preference`.

If you're providing moving media to developers, also provide a non-moving variant that they can use in the media query.

The recommended way is to only add motion if the user hasn't set a preference:

```
@media (prefers-reduced-motion: no-preference) { .moving-element { animation: bounce 500ms; } }
```

Useful Links

WCAG 2.2 Techniques - Using CSS prefers-reduced-motion to prevent motion

Respecting users' motion preferences

MDN - prefers-reduced-motion

Presentations

Inclusion Toolkit

What is it for?

Most people will have to give a presentation at some point. Presentations are used to convey information, so it's very important that everyone can get the information being provided.

How to support it

- Attach your slides to the meeting invite, so everyone can see the slides in advance
- Include alt text on all images in your slides
- Consider your font. Make sure it's at least 18 point in size, and use a sans serif font
- Use left-aligned text, as it's easier to read for many people (or right-aligned text for languages that are read from the right)
- Don't use images of text
- Check your colour contrast
- If you're including links, make sure the link text is descriptive
- Ensure that colour isn't your only way of conveying information

Useful Links

Making presentations more accessible

NHS England: Making meetings accessible, presentations section

QR Codes

Inclusion Toolkit

What is it for?

A QR (Quick Response) code is a type of two-dimensional barcode. It's a machine-scannable image that a smartphone camera can read to link a physical item to a website or app.

QR codes create significant barriers if not implemented carefully. These barriers include:

- Users who are blind or have low vision cannot easily locate or scan a QR code
- A standard code provides no information about its destination before being scanned

How to support it

- Always include a short, readable uniform resource locator (URL) and a clear description of the link's purpose next to the code
- Use high-contrast colours (dark on light), a large size (at least 2.5 by 2.5 centimetres), and place it at a height reachable by all users

If you can avoid using a QR code, you should. Alternatives include short URLs, near-field communication (NFC) tags and Accessible QR (AQR) codes.

Useful Links

Accessibility and QR codes

What is an Accessible QR (AQR) code?

Making QR codes more accessible

Are QR Codes Accessible for People with Disabilities?

Accessible QR Codes – The Ultimate Guide

Screen Reader Only Styling

Inclusion Toolkit

What is it for?

Screen reader only styling ensures that everyone who uses a website can access the important information it contains. In most cases, this is achieved with semantics, alt text, and aria labels. However, in some cases, we want something to only be 'seen' by screen readers.

For instance, a blog might have several 'Read more' links. You can add words to each, such as 'about accessible meetings', that are hidden visually but read out by screen readers.

How to support it

You can add the code on this page into your CSS, Leaner Style Sheets (LESS), Syntactically Awesome Style Sheets (SASS), or any other CSS-based styling.

When you're using things like `display: none;` consider whether a screen reader user would benefit from the thing you're hiding. That's a great time to use this styling!

```
.sr-only { position: absolute;
  width: 0.0625rem; border: 0;
  height: 0.0625rem; padding: 0;
  margin: -0.0625rem;
  overflow: hidden;
  clip: rect(0 0 0 0);
  white-space: nowrap; }
```

Useful Links

WebAIM - Invisible content

CSS Tricks - Inclusively Hidden

Screen Readers

Inclusion Toolkit

What is it for?

Screen readers are pieces of software that announce content audibly. This allows a computer to read text, images, icons, and much more. Screen readers are used if someone does not have enough vision to be able to view content from a screen.

They're especially important for people with visual impairments, learning difficulties or cognitive disabilities, non-native speakers, and people who prefer to listen to content instead of reading it visually.

How to support it

Most devices come with a screen reader installed as standard: VoiceOver on Mac and iPhone, Narrator on Windows, and TalkBack on most Android devices.

- Decorative images and icons aren't read out
- All images are understandable using the alt text
- The order of the page makes sense
- Everything on the page is fully functional using a screen reader

Useful Links

An Introduction to Screen Readers

RNIB: Screen reading software

Shifting Left

Inclusion Toolkit

What is it for?

Imagine baking a cake, pulling it out of the oven, and realising you forgot the sugar. It's much easier to just add the sugar at the beginning.

Shifting left means moving accessibility checks to the very beginning of the project (the left side), instead of waiting until the end to test it. Finding an accessibility bug during the design phase is very cheap and fast to fix.

How to support it

- Designers can check colour contrast and plan out where keyboard focus will go before sending pictures to the developers
- Developers can plan out which Semantic HTML tags to use before writing any complex code
- Product Managers can make sure accessibility is listed as a requirement for the feature to be considered Done

This early planning saves a lot of time and money at the end.

Useful Links

Deque: Shift Left Accessibility Testing

Shift-left accessibility: Embedding inclusion early in agile projects

Social Media: Alt Text

Inclusion Toolkit

What is it for?

On social media, images often hold the core message or punchline. Alt text ensures that context isn't lost for anyone by serving a few vital purposes:

- Allows screen readers to describe the visual content to visually impaired users
- Displays text descriptions when poor signal or low data prevents images from loading
- Helps users with cognitive disabilities better understand complex graphics or memes
- Boosts discoverability by helping platform algorithms understand and index your posts

How to support it

In workplace chat tools like Slack, upload the image, click ‘Edit file details’, and fill in the description.

In PowerPoint, right-click the image, then ‘View Alt Text’. In Google Slides, use ‘Format options’, then ‘Alt text’.

On LinkedIn, add the image and click ‘ALT’ at the bottom right.

On Facebook, add the image, open its menu (the button with three dots), and select ‘Edit alt text’.

On X (Twitter), add the image and click the ‘ALT’ box at the bottom.

Useful Links

Alternative text on social media

Alternative text in action

Social Media and Alternative Text: A Reflection

Social Media: Custom Fonts

Inclusion Toolkit

What is it for?

People often use generator websites to make their text look unique (for example, 𝒸𝓊𝓇𝓈𝒾𝓋𝑒 [the word cursive, in fancy script letters] or 𝕓𝕠𝕝𝕕 [the word bold, in outlined letters]). However, these are not real fonts. They are mathematical symbols from the Unicode standard designed for equations, not language.

When a screen reader encounters this text, it does not read the word. It reads the mathematical description of every single character. 𝐻𝑒𝓁𝓁𝑜 ("Hello") becomes "Mathematical script capital H, mathematical script small e..." which makes the message impossible to understand.

How to support it

The kindest thing you can do is avoid these generators entirely for your display name, profile description, and post text.

- Use the default keyboard font provided by the platform. This ensures your text is readable and searchable.
- Since you cannot use bold or italics on most apps, capitalise the first letter of key words (not whole words) instead.

Useful Links

Accessible Social: Just Not My Type, on why fancy text generators cause problems

Social Media: Emojis

Inclusion Toolkit

What is it for?

Emojis add personality and tone to our messages, but for a screen reader user, they are read aloud as literal text descriptions. For example, a simple smiley face is read out as 'Smiling face with open mouth'.

When emojis are used excessively or incorrectly, this feature can become overwhelming. If you string ten 'Clapping hands' emojis together, the user must listen to the phrase 'Clapping hands' repeated ten times in a row. Similarly, if you use an emoji as a bullet point, the user hears the emoji name before every single item on the list, which distracts from the content.

How to support it

To keep your content fun but readable, follow these best practices:

- Use them sparingly. One or two emojis are usually enough to convey a mood
- Put them at the end. Place emojis at the end of a sentence or paragraph so they do not interrupt the flow of reading
- Never replace a word with an emoji (for example, writing 'I want [pizza emoji]')
- Check the emoji description. Use a site like Emojipedia to check exactly how your chosen emoji will be described to a blind user

Useful Links

Emojipedia

Accessible Social - Emoji

The Good, The Bad, And The Ugly: Emojis and Accessibility

Social Media: Hashtags

Inclusion Toolkit

What is it for?

Imagine trying to read a sentence where all the spaces have been removed. It would look like a jumble of letters that takes real effort to decipher. That is exactly what a screen reader encounters when it reads a hashtag written entirely in lowercase.

If you write #superbowl, a screen reader might read it as "Superb Owl" rather than "Super Bowl". By capitalising the first letter of each word, we give the technology a clear signal of where one word ends and the next begins. This ensures the hashtag is pronounced correctly and is easier for everyone to read visually.

How to support it

Make hashtags accessible with a few simple adjustments.

- Always capitalise the first letter of every word, like '#SocialMediaMarketing'. This helps screen readers pronounce words correctly and improves clarity
- Keep your hashtags short. Combining too many words makes them difficult to process, even with capital letters
- Place hashtags at the very end of your post. Don't put them in the middle of a sentence, as this disrupts the reading flow for assistive technology users

Useful Links

Accessible Social

RNIB - How to make your social media accessible

Hootsuite - Inclusive Design for Social Media

Tap Targets

Inclusion Toolkit

What is it for?

Make sure interactive elements, like buttons or links, are large enough and have enough space around them. This makes them easy to press without touching other elements by mistake.

This can remove barriers for users with mobility impairments (such as hand tremors), users on touchscreens or in environments where they are exposed to shaking, users who access a device using one hand, and users with low vision.

How to support it

Web Content Accessibility Guidelines (WCAG) 2.2 needs tap targets of at least 24 by 24 pixels, or enough space around them (Level AA). Level AAA needs 44 by 44 pixels.

Google's Material Design recommends at least 48 by 48 pixels, with a gap of at least 8 pixels between them.

An icon inside a link may only be 24 by 24 pixels. You can add padding to the link to bring the tap target size up to 48 by 48 pixels.

Useful Links

Understanding SC 2.5.8: Target Size (Minimum)

How to use typography, touch targets, and spacing for better accessibility

Accessible tap targets cheat sheet

Temporary & Situational Disabilities

Inclusion Toolkit

What is it for?

Accessibility is often associated with permanent disabilities. But disability exists on a much wider spectrum.

A temporary disability will go away over time, like a broken arm or an eye infection that makes screens painful to look at.

A situational disability occurs when your environment creates the same barriers as a disability. For example, reading a phone in bright sunlight, or following audio in a loud pub.

How to support it

- Designers can ask, 'what if someone is doing this in harsh sunlight?' when checking colour contrast
- Developers can add captions to all video content
- Product Managers can include situational scenarios in user stories, like 'a user completing this task while commuting'

Designing for someone holding a baby also means designing for someone with a broken arm, and for someone with a permanent upper limb disability. One solution, three problems solved.

Useful Links

Microsoft Inclusive Design

W3C: Diverse Abilities and Barriers

Text Alternatives for Complex Images

Inclusion Toolkit

What is it for?

Standard 'alt text' is great for simple photos, but it often fails for complex data like charts, graphs, flowcharts, or infographics. A single sentence cannot convey all the information provided. People using screen readers (or anyone who struggles to process dense visual data) must be able to access the same information as everyone else.

How to support it

It needs 2 steps, because a short description is not enough

- Use the standard 'alt text' to briefly identify the image (for example, 'Bar chart showing revenue')
- Provide the full data in a text format nearby

For charts and graphs, place a properly formatted data table immediately below the image.
For infographics, provide a full text transcript or a link to a page that outlines all the information in the same order.
For diagrams, write a structured list or description that walks the user through the flow.

Useful Links

WAI - Complex Images

WebAIM - Accessible Images

Time Limits

Inclusion Toolkit

What is it for?

Time limits refer to when a page sets a time limit on doing a certain action. This is often seen while filling in forms, or whilst making a purchase.

To make time limits accessible, Web Content Accessibility Guidelines (WCAG) 2.2 has a Level A rule. Users must be able to turn a time limit off, make it at least 10 times longer, or get a warning and 20 seconds to extend it.

Time limit considerations are most important for users with motor skills impairments, users with cognitive difficulties and screen reader users.

How to support it

Ensure that any time limits you use can be manually extended. Also ensure that the implementation of those time limits isn't done in a scary or anxiety-inducing manner.

Useful Links

Access Guide: Time limits

Understanding SC 2.2.1: Timing Adjustable

Web Accessibility Tips: Give People Enough Time

Tooltips & Popovers

Inclusion Toolkit

What is it for?

Tooltips and popovers give users extra information on demand, but they often fail accessibility by being:

- Mouse-only. They don't appear for keyboard users (focus)
- Silent. Screen readers don't announce the new content
- Hard to close. Users get 'stuck', or the tooltip vanishes too fast

It's important to note that popovers and modals are not the same thing. A modal demands your full attention, while a popover just offers extra information

How to support it

They must work for keyboard, mouse, and screen reader users.

- Simple tooltips (read-only information) should show on hover and focus, stay visible, close with Escape, and use `aria-describedby` to link the trigger to the tooltip so screen readers announce it
- Complex popovers (with buttons or links inside) should trigger on click (not hover), manage keyboard focus, and be easy to close

Useful Links

WAI-ARIA Authoring Practices - Tooltip Pattern

Touch Gestures

Inclusion Toolkit

What is it for?

Imagine trying to open a heavy door, but the rule is you can only use your little fingers. Some people have shaky hands, or use a special typing wand held in their mouth. For them, doing a pinch-to-zoom or a three-finger swipe on a screen is just as hard. If a map app only lets you zoom in by pinching, those users can't zoom in at all.

How to support it

Always provide a simple, single-finger tap option for every complex action.

- If a map requires pinching to zoom, include standard tap buttons like '+' and '-'
- If a panel requires swiping, add a visible menu button to tap instead
- Ensure no vital action relies on twisting, pinching, or multi-finger movements
- Keep complex gestures as handy shortcuts, but don't make them the only way to get things done

Useful Links

W3C - Understanding Pointer Gestures

WebAIM - Motor Disabilities

Zooming

Inclusion Toolkit

What is it for?

Some people need to make the text and images on a website bigger to see them clearly. Pages must support this. When a user zooms in, the content should automatically rearrange itself to fit the screen. This is often called 'reflow'. A good page will reflow so that users never have to scroll from side to side to read a single sentence. This is very important for creating a positive experience for people with low vision.

All modern browsers support zooming. It's up to developers and designers to build layouts that still work when people zoom in.

How to support it

On a computer: hold the Control (Ctrl) key, or the Command (Cmd) key on a Mac, and press +. Check at 200%, then zoom to 400%. Can you read everything without scrolling left and right?

On a mobile: put two fingers on your screen, close to each other, and spread them apart.

Note: The 'pinch-to-zoom' feature must never be blocked. Disabling zoom creates a major barrier for accessibility.

Useful Links

Adrian Roselli - Don't Disable Zoom

CSS-Tricks - Accessible Font Sizing, Explained

WebAIM - Responsive Design and Reflow

Glossary of abbreviations (1 of 2)

Inclusion Toolkit

a.k.a.: Also known as

A11y: Accessibility (a numeronym: 11 letters between the 'a' and the 'y')

AA and AAA: WCAG conformance levels. AA is the middle level; AAA is the highest

ADHD: Attention deficit hyperactivity disorder

Alt text (ALT): Alternative text

AMOLED: Active-matrix organic light-emitting diode (a type of screen)

AQR: Accessible QR code

ARIA: Accessible Rich Internet Applications

ASCII: American Standard Code for Information Interchange

AT: Assistive technology

CSS: Cascading Style Sheets

e.g.: For example

etc.: And so on

HMRC: His Majesty's Revenue and Customs

HTML: HyperText Markup Language

i.e.: That is

LESS: Leaner Style Sheets (a CSS preprocessor)

Glossary of abbreviations (2 of 2)

Inclusion Toolkit

MDN: MDN Web Docs (originally the Mozilla Developer Network)

NFC: Near-field communication

OLED: Organic light-emitting diode (a type of screen)

PAC: PDF Accessibility Checker

PDF: Portable Document Format

pt: Points (a unit of font size)

px: Pixels

Q3: The third quarter of the year

QR: Quick Response (as in QR code)

RNIB: Royal National Institute of Blind People

SASS: Syntactically Awesome Style Sheets (a CSS preprocessor)

SC: Success criterion (a single testable WCAG requirement)

SEO: Search engine optimisation

TV: Television

URL: Uniform Resource Locator (a web address)

W3C: World Wide Web Consortium

WAI: Web Accessibility Initiative

WCAG: Web Content Accessibility Guidelines

Glossary of terms (1 of 4)

Inclusion Toolkit

Ableist language: Words that use disability as an insult or a negative image, such as 'lame' or 'blind spot'

Accessibility debt: Accessibility problems that build up when accessibility is left until later

Accessibility overlay: A third-party add-on that claims to fix a website's accessibility problems automatically

Alt text: A short written description of an image, read out by screen readers

Assistive technology: Tools that help disabled people use a device, such as screen readers, magnifiers, switches, and voice control

Autocomplete / autofill: When a browser or password manager fills in saved details for you

Biometrics: Using a fingerprint or face scan to sign in

Breadcrumbs: A trail of links showing where a page sits on a website, such as Home, then Phones, then iPhone

CamelCase: Writing each word in a hashtag with a capital letter, such as #SuperBowl

Captions: Text on screen showing the speech and important sounds in a video

Carousel: A banner that moves through a set of images or messages

Cognitive: To do with thinking, memory, attention, and understanding

Glossary of terms (2 of 4)

Inclusion Toolkit

Colour contrast ratio: How different two colours are in brightness. A higher number is easier to read, such as 4.5 to 1

Component library / design system: A shared set of ready-made, tested parts, such as buttons and forms, used to build products

Decorative image: An image that adds nothing to the meaning, so screen readers can skip it

Focus: Which element is currently active and will receive keyboard input

Focus state: How an element looks when it has focus, such as a visible outline

Focus trap: When keyboard users can't tab away from part of a page

Gesture: A touch action, such as a swipe, pinch, or multi-finger tap

Hover state: How an element looks when a mouse pointer is over it

Identity-first language: Putting the identity first, such as 'an autistic person'

Interactive element: Anything you can click, tap, or type into, such as a link, button, or form field

Landmark: A labelled area of a web page, such as the navigation or main content, that screen reader users can jump to

Luminance: How bright a colour looks

Glossary of terms (3 of 4)

Inclusion Toolkit

Magic link: A one-time sign-in link sent by email

Media query: A rule in CSS that changes how a page looks based on the device or the user's settings

Modal: A pop-up window that appears on top of a page and must be closed before you go back to the page

Motor impairment: A disability that affects movement, such as tremors or limited use of the hands

Neurodiversity: The natural differences in how people's brains work, including autism, ADHD, dyslexia, and dyspraxia

One-time code: A short code sent by text or app that you use once to sign in

Person-first language: Putting the person first, such as 'a person with vision loss'

Persona: A made-up but realistic person used to test whether a product works for different people

Photosensitive epilepsy: Epilepsy where seizures can be triggered by flashing or flickering light

Reflow: When a page rearranges itself to fit the screen, so you don't need to scroll sideways

Screen magnifier: Software that enlarges part of the screen

Screen reader: Software that reads out what is on the screen

Glossary of terms (4 of 4)

Inclusion Toolkit

Semantic HTML: Using the HTML element that matches the content's meaning, such as `<button>` for a button

Shifting left: Thinking about accessibility at the start of a project rather than the end

Skip link: A link at the top of a page that lets keyboard and screen reader users jump past the navigation

Switch device: A single button or similar device used to control a computer, for people with limited movement

Tagged PDF: A PDF with hidden structure (headings, lists, reading order) that screen readers can use

Tap target: The area of a button or link you can tap or click

Tooltip / popover: A small box of extra information that appears next to an element

Transcript: A written version of everything said, and important sounds, in audio or video

Vestibular disorder: An inner-ear condition that affects balance, which motion on screen can make worse

Voice control: Controlling a device by speaking commands